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TECH 4272

Week 5 Deliverables

1. Production Environment Setup Document production deployment:

```
ubuntu@FOSSBilling: ~  
GNU nano 6.2 deployment_guide.md  
# Production Deployment Guide: Cloud Infrastructure  
  
**Role:** Infrastructure Lead  
**Document Version:** 1.0  
**Status:** Ready  
  
## 1. Architecture Overview  
This document outlines the standard operating procedure for deploying our high-availability (HA) VPS hosting infrastructure. The environment  
  
## 2. Hardware Allocation  
  
### Proxmox Host (testhost)  
- **CPU:** 12 Cores  
- **RAM:** 46.89 GB  
- **Storage:** 94 GB  
  
### Virtual Machine Inventory  
| VM ID | Hostname | Role | CPU | RAM | Storage |  
|-----|-----|-----|-----|-----|-----|  
| 100 | 'fossbilling' | Customer Portal & Database | 4 vCPU | 8 GB RAM | 20 GB |  
| 200 | 'haproxy-lb' | Incoming Traffic Router | 2 vCPU | 2 GB RAM | 12 GB |  
| [ID] | 'monitor-01' | Grafana / Prometheus Metrics | 2 vCPU | 4 GB RAM | 20 GB |  
  
## 3. Network & Security Configuration  
  
### Network Segmentation  
The infrastructure utilizes a dual-network approach to separate public management traffic from internal application traffic:  
* **Public Management:** Attached to 'vmbr0' (Allows external access to Proxmox GUI).  
* **Private Network ('localnetwork'):** Attached to 'vmbr1' (Handles all internal communication between VMs without exposing them to the  
  
### Firewall (UFW) Rules  
The following strict 'ufw' rules are applied to the core FOSSBilling VM to ensure minimal attack surface:  
  
'''bash  
# Default deny incoming, allow outgoing  
sudo ufw default deny incoming  
sudo ufw default allow outgoing  
  
# Allow SSH for administration  
sudo ufw allow 22/tcp  
  
# Allow Web Traffic (HTTP/HTTPS)  
sudo ufw allow 80/tcp  
sudo ufw allow 443/tcp  
'''  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo       M-A Set Mark  M-G Copy  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line  M-E Redo      M-G Copy
```

2. Backup and Recovery Testing

Time Required for each backup;

FOSSBilling – 1 Minute

Restore - 48 Seconds

VMs Backup – 2 Minutes

Restore – 33 Seconds

Restore Database from Backup

Current Status:

5 Active

0 Suspended

0 Canceled

5 Total

Clients

+ New Client

Groups

+ New Group

Enter search text...

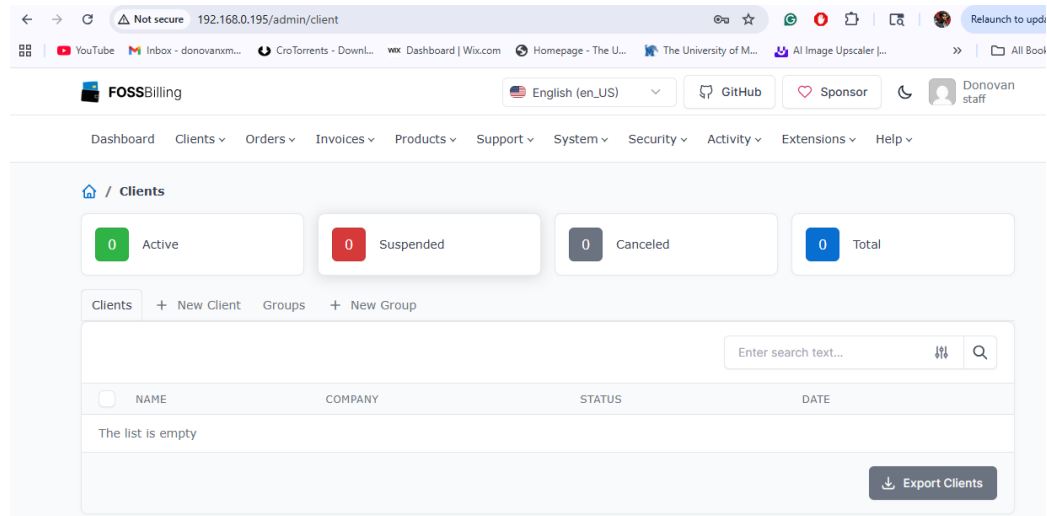
🔍

☐	NAME	COMPANY	STATUS	DATE		
☐	Jordan jordansteele1316@gmail.com	-	● Active	Apr 13, 2026, 11:36 PM	✎	🗑
☐	Jordan2 steelejordan545@gmail.com	-	● Active	Apr 6, 2026, 10:35 PM	✎	🗑
☐	Faruk faruk@gmail.com	-	● Active	Apr 1, 2026, 8:11 PM	✎	🗑
☐	Jordan jordan@gmail.com	-	● Active	Apr 1, 2026, 8:06 PM	✎	🗑
☐	Ahmed ahmed@gmail.com	-	● Active	Apr 1, 2026, 7:44 PM	✎	🗑

Backup:

```
ubuntu@FOSSBilling: ~  
Last login: Thu Apr 23 01:25:25 2026 from 192.168.0.190  
ubuntu@FOSSBilling:~$ ls -la /var/www/fossbilling  
total 132  
drwxrwxr-x 8 www-data www-data 4096 Apr 22 21:23 .  
drwxr-xr-x 4 root root 4096 Apr 1 18:09 ..  
-rwxrwxr-x 1 www-data www-data 1561 Sep 7 2025 .htaccess  
-rwxrwxr-x 1 www-data www-data 11454 Sep 7 2025 LICENSE  
-rwxrwxr-x 1 www-data www-data 6063 Sep 7 2025 README.md  
-rwxrwxr-x 1 www-data www-data 6624 Sep 7 2025 config-sample.php  
-rw-r--r-- 1 www-data www-data 1646 Apr 1 18:51 config.old.php  
-rw-r--r-- 1 www-data www-data 1700 Apr 22 21:23 config.php  
-rwxrwxr-x 1 www-data www-data 2046 Sep 7 2025 console.php  
-rwxrwxr-x 1 www-data www-data 992 Sep 7 2025 cron.php  
drwxrwxr-x 6 www-data www-data 4096 Apr 22 21:23 data  
-rwxrwxr-x 1 www-data www-data 23036 Sep 7 2025 di.php  
-rwxrwxr-x 1 www-data www-data 2882 Sep 7 2025 index.php  
-rwxrwxr-x 1 www-data www-data 1846 Sep 7 2025 ipn.php  
drwxrwxr-x 9 www-data www-data 4096 Sep 7 2025 library  
-rwxrwxr-x 1 www-data www-data 10775 Sep 7 2025 load.php  
drwxrwxr-x 34 www-data www-data 4096 Sep 7 2025 locale  
drwxrwxr-x 46 www-data www-data 4096 Apr 8 23:16 modules  
-rwxrwxr-x 1 www-data www-data 1329 Sep 7 2025 rewrite-missing.html  
-rwxrwxr-x 1 www-data www-data 615 Sep 7 2025 robots.txt  
drwxrwxr-x 4 www-data www-data 4096 Sep 7 2025 themes  
drwxrwxr-x 37 www-data www-data 4096 Sep 7 2025 vendor  
ubuntu@FOSSBilling:~$ sudo sh -c 'mysqldump --all-databases > full_backup.sql'  
ubuntu@FOSSBilling:~$ ls -lh full_backup.sql  
-rw-r--r-- 1 root root 41M Apr 25 18:39 full_backup.sql  
ubuntu@FOSSBilling:~$
```

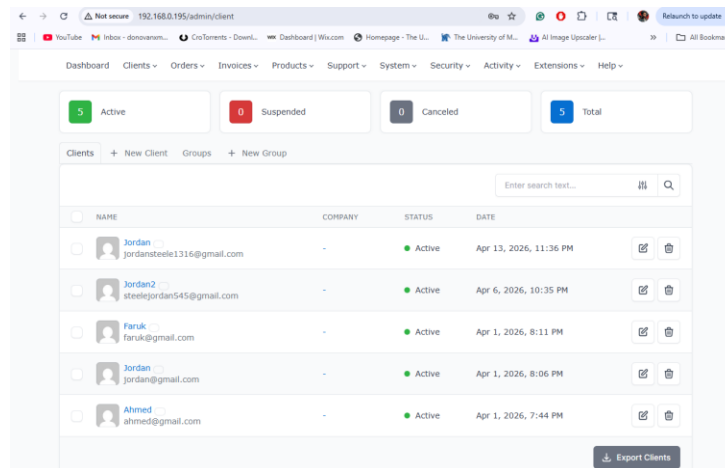
Disaster Scenario: All Customers Deleted From Database



The screenshot shows the FOSSBilling dashboard at the URL 192.168.0.195/admin/client. The top navigation bar includes links for Dashboard, Clients, Orders, Invoices, Products, Support, System, Security, Activity, Extensions, and Help. The main content area displays the 'Clients' section with four summary cards: Active (0), Suspended (0), Canceled (0), and Total (0). Below these cards are links for 'Clients', '+ New Client', 'Groups', and '+ New Group'. A search bar with the placeholder 'Enter search text...' is present. The table below the search bar is empty, displaying the message 'The list is empty'. An 'Export Clients' button is located at the bottom right of the table area.

NAME	COMPANY	STATUS	DATE
The list is empty			

Recovery:



The screenshot shows the FOSSBilling dashboard after recovery. The summary cards now show: Active (5), Suspended (0), Canceled (0), and Total (5). The table below the search bar is populated with five client records, each with a checkbox, profile picture, name, email, company, status, date, and action icons.

NAME	COMPANY	STATUS	DATE
☐ Jordan jordansteele1316@gmail.com	-	Active	Apr 13, 2026, 11:36 PM
☐ Jordan2 steelejordan545@gmail.com	-	Active	Apr 6, 2026, 10:35 PM
☐ Faruk faruk@gmail.com	-	Active	Apr 1, 2026, 8:11 PM
☐ Jordan jordan@gmail.com	-	Active	Apr 1, 2026, 8:06 PM
☐ Ahmed ahmed@gmail.com	-	Active	Apr 1, 2026, 7:44 PM

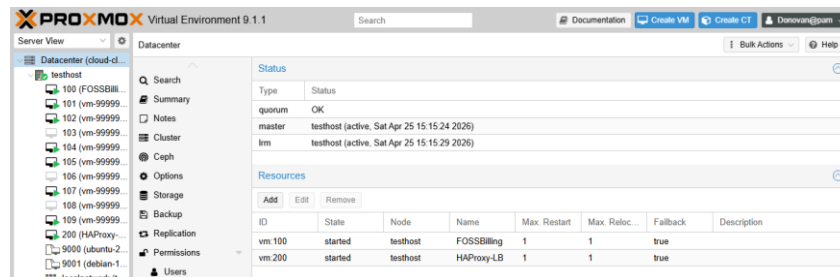
Time Required for database recovery: About 3 Minutes

3. Capacity Planning

```
ubuntu@FOSSBilling: ~  
GNU nano 6.2 capacity_calculator.py  
def calculate_capacity(servers):  
    """Calculate total customer capacity"""  
    total_cpu = sum(s['cpu'] for s in servers)  
    total_ram = sum(s['ram'] for s in servers)  
    total_disk = sum(s['disk'] for s in servers)  
  
    # Account for overhead (20%)  
    usable_cpu = total_cpu * 0.8  
    usable_ram = total_ram * 0.8  
    usable_disk = total_disk * 0.8  
  
    # Calculate VMs per plan  
    plans = {  
        'micro': {'cpu': 1, 'ram': 1, 'disk': 20},  
        'small': {'cpu': 2, 'ram': 2, 'disk': 40},  
        'medium': {'cpu': 4, 'ram': 4, 'disk': 80},  
        'large': {'cpu': 8, 'ram': 8, 'disk': 160}  
    }  
  
    capacity = {}  
    for plan_name, specs in plans.items():  
        max_by_cpu = usable_cpu / specs['cpu']  
        max_by_ram = usable_ram / specs['ram']  
        max_by_disk = usable_disk / specs['disk']  
  
        capacity[plan_name] = int(min(max_by_cpu, max_by_ram, max_by_disk))  
  
    return capacity  
  
servers = [  
    {'name': 'testhost', 'cpu': 12, 'ram': 47, 'disk': 94},  
]  
  
print(calculate_capacity(servers))  
  
ubuntu@FOSSBilling:~$ python3 capacity_calculator.py  
{'micro': 3, 'small': 1, 'medium': 0, 'large': 0}  
ubuntu@FOSSBilling:~$
```

4. Disaster Recovery Drill

Before Disaster

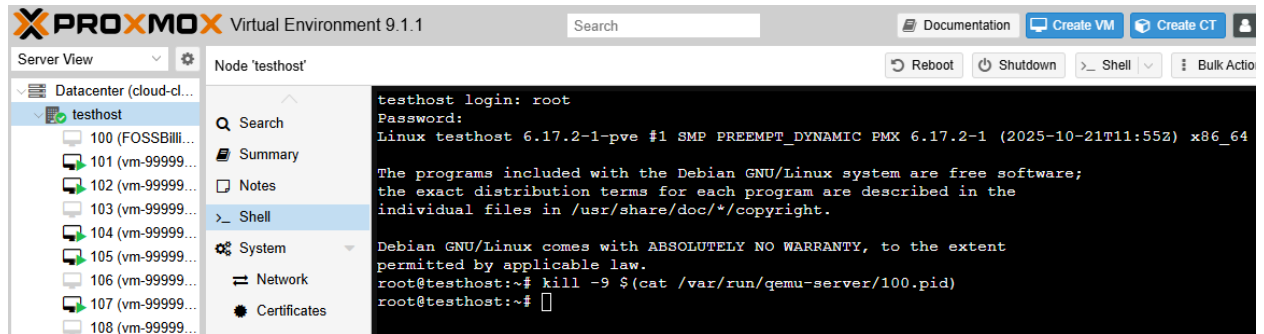


The screenshot shows the Proxmox VE 9.1.1 interface. The left sidebar displays the 'Datacenter (cloud-d...)' tree with various VMs listed. The main panel shows the 'Status' and 'Resources' sections. The 'Status' section includes a table with columns 'Type' and 'Status', showing 'quorum' as 'OK', 'master' as 'testhost (active, Sat Apr 25 15:24:2026)', and 'lrm' as 'testhost (active, Sat Apr 25 15:29:2026)'. The 'Resources' section includes a table with columns 'ID', 'State', 'Node', 'Name', 'Max. Restart', 'Max. Reloc.', 'Fallback', and 'Description', showing two VMs: 'vm 100' (started, testhost, FOSSBilling) and 'vm 200' (started, testhost, HAProxy-LB).

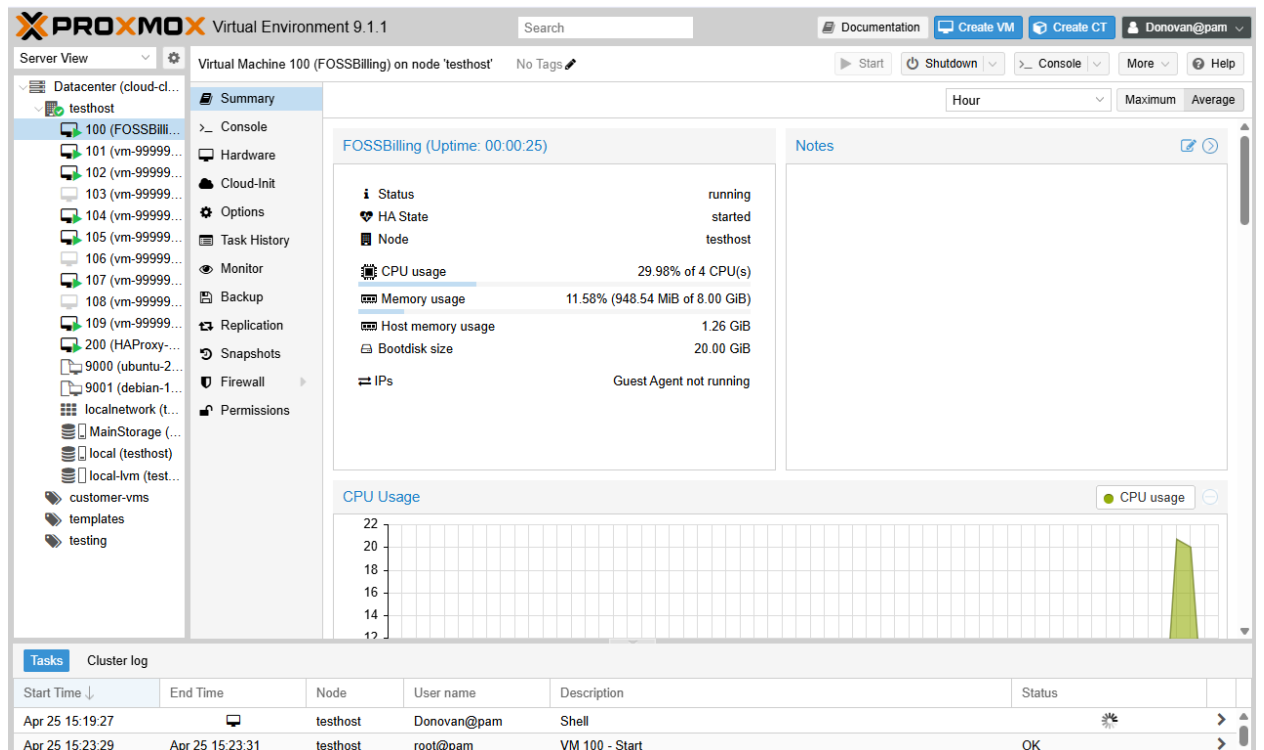
Type	Status
quorum	OK
master	testhost (active, Sat Apr 25 15:24:2026)
lrm	testhost (active, Sat Apr 25 15:29:2026)

ID	State	Node	Name	Max. Restart	Max. Reloc.	Fallback	Description
vm 100	started	testhost	FOSSBilling	1	1	true	
vm 200	started	testhost	HAProxy-LB	1	1	true	

During



After Recovery



The most surprising lesson learned during the Recovery is the speed of the Proxmox HA. The system detected the process crash, reallocated its resources, and fully rebooted the FOSSBilling VM faster than a web browser could be manually refreshed. Also during the first attempt to crash the VM using my standard user account, the hypervisor immediately blocked the command with a “Permission denied” error.